

1. Suppose there are 200 observations and we perform **10-fold cross-validation**.

For each fold, KNN is trained on 180 observations and evaluated on 20.

Question: Across the entire 10-fold CV procedure:

1. How many times is each observation used for validation?
2. How many times is each observation used for training?
3. How many models are trained?

1. a) Each observation is used exactly once for validation

b) Each observation is used 9 times for training
(Each observation is excluded from training
when its own fold is a validation set)

c) For each candidate k , there is one model
per fold/iteration. $\therefore$ There are 10 models
for each candidate k .

2. You perform 5-fold CV and get

(k)	Mean CV Accuracy
1	91%
3	93%
5	93%
7	92%

Suppose the standard deviations of the five fold accuracies are:

(k)	Mean	SD
1	91%	1%
3	93%	2%
5	93%	0.5%
7	92%	1%

Which k would you prefer between $k = 3$ and $k = 5$, and why?

2. $k=5$

$k=3$ and $k=5$ have same mean accuracy
but $k=5$ has less std. dev. This indicates
that $k=5$ is more stable and consistent

3. Suppose:

(k)	Test Accuracy
5	91.2%
7	91.3%
9	91.2%
11	90.9%

Would you necessarily choose $k = 7$? Why/Why not?

3. No. Not necessarily.

The difference between $k=7$ and $k=5$ or $k=9$ is just 0.1% points.

This could be due to the particular test set we chose. Can be a chance occurrence.

Correct procedure:

1. Use cross validation to choose k .
2. Run the selected model on test-data once k is finalized. (ONLY ONCE)

4. Suppose you are comparing two choices of k

$k = 3$

Fold	Accuracy
1	95%
2	95%
3	95%
4	75%
5	75%

$k = 7$

Fold	Accuracy
1	89%
2	90%
3	91%
4	89%
5	90%

Which model appears more stable and why?

4. Compare the variations in fold accuracies!

For $k=3$, varies between 75% to 95%.

For $k=7$, varies between 89% to 91%.

This makes $k=7$ more stable

5. Suppose cross validation gives:

(k)	CV Accuracy
1	88%
3	91%
5	94%
7	93%
9	92%

You choose $k = 5$ and evaluate the model on the test set and get 90% accuracy. Your colleague says “Since $k = 7$ might have done better on the test set, let's try $k = 7$ and see.”

Is this a good or bad idea? Why?

5. This is generally a bad idea.
 $k=5$ was chosen because of its high
CV accuracy. It was then tested on

the test data.

The test data has done its job: Provided
an unbiased estimate of our model performance.

If we test $k=7$ on test data, we are
implicitly using the test set to make modeling
decisions and it is no longer truly
untouched / independent.

To truly reconsider k , go back to CV step.
Select different k for k -fold CV and
re-run the process

6. An airport security system uses ML to identify potentially dangerous passengers.

Suppose the system has:

- 100 dangerous passengers
- 10,000 safe passengers

The system catches 95 of the 100 dangerous passengers but incorrectly flags 500 safe passengers.

- What is recall?
- What is precision?
- What metric should be prioritized?

6.

Confusion matrix

		Predicted	
Actual	+	TP = 95	FN = 5
	-	FP = 500	TN = 9500

$$\text{Recall} = \frac{TP}{TP + FN} = 95\%$$

$$\text{Precision} = \frac{TP}{TP + FP} \approx 16\%$$

Recall would be a better metric because missing a genuinely dangerous person can have dire consequences.

However, 500 FP is a lot, so additional considerations need to be made

7. Imagine you're building a search engine.

A user searches: "machine learning textbooks"

The search engine returns 20 results. Only 15 are actually relevant.

- Which metric is directly measuring how useful those 20 returned results are?

7.

we are asking,
out of the books selected as relevant, how
many / what % is actually relevant?

$$\frac{TP}{TP + FP} = \text{Precision} = \frac{15}{20}$$

∴ The metric to be used is "Precision"

8. Two models are being considered for detecting a disease:

Model	Precision	Recall
A	95%	60%
B	80%	95%

The disease is serious, and the consequences of **missing a sick patient** are much greater than the consequences of sending a healthy patient for additional testing.

Which model would you choose and why?

8. Model B since it has higher Recall.